

Homework 6- Solutions

Problem 1. Let X and Y be two random variables, whose joint PMF $p_{X,Y}(x, y)$ is given by the following table:

$x = 1$	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{1}{8}$
$x = 0$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{4}$
	$y = -1$	$y = 0$	$y = 1$

Using this table, evaluate the following quantities.

(1.1) $Var(X|Y = 0)$.

(1.2) $p_{X|Y}(1|A)$, where $A = \{Y^2 > 0\}$

Solution.

(1.1) $X|(Y = 0) \sim \text{Bern}(p)$ where $p = P(X = 1|Y = 0)$. First we need to calculate

$$p_{X|Y}(1|0) = \frac{P(X = 1, Y = 0)}{P(Y = 0)} = \frac{\frac{1}{16}}{\frac{3}{16}} = \frac{1}{3}.$$

Now from The formula of variance of a Bernoulli distribution we have

$$Var(X|Y = 0) = p(1 - p) = \frac{1}{3}\left(1 - \frac{1}{3}\right) = \frac{2}{9}$$

$$(1.2) \quad p(X = 1|A) = \frac{P(X = 1, A)}{P(A)} = \frac{\frac{5}{16}}{\frac{13}{16}} = \frac{5}{13}$$

Problem 2. Consider a joint probability density function $f(x, y)$ is given by

$$\begin{cases} \frac{16y}{x^3} & x > 2, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find $E[X]$ and $E[Y]$.

(ii) Find $Cov(X, Y)$

(iii) Are X and Y correlated? Why/Whynot?

First we should find the marginal pdf of X and the marginal pdf of Y .

$$f_X(x) = \begin{cases} \int_0^1 \frac{16y}{x^3} dy = \frac{8y^2}{x^3} \Big|_0^1 = \frac{8}{x^3} & x > 2 \\ 0 & x \leq 2 \end{cases} \quad (1)$$

$$f_X(x) = \begin{cases} \frac{8}{x^3} & x > 2 \\ 0 & x \leq 2 \end{cases} \quad (2)$$

$$E[X] = \int_2^\infty x \frac{8}{x^3} dx = \int_2^\infty 8x^{-2} dx = -\frac{8}{x} \Big|_2^\infty = 0 + \frac{8}{2} = 4$$

$$f_Y(y) = \begin{cases} \int_2^\infty \frac{16y}{x^3} dx = -\frac{8y}{x^2} \Big|_2^\infty = 2y & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

$$f_Y(y) = \begin{cases} 2y & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

$$E[Y] = \int_0^1 2y^2 dy = \frac{2y^3}{3} \Big|_0^1 = \frac{2}{3}$$

$$\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$\begin{aligned}
E[XY] &= \int_0^1 \int_2^\infty xy \frac{16y}{x^3} dx dy \\
&= \int_0^1 \int_2^\infty 16y^2 x^{-2} dx dy \\
&= \int_0^1 16y^2 dy \int_2^\infty x^{-2} dx \\
&= \frac{16y^3}{3} \Big|_0^1 \cdot \frac{-1}{x} \Big|_2^\infty = \frac{16}{3} \cdot \frac{1}{2} = \frac{8}{3}
\end{aligned}$$

$$Cov(X, Y) = E[XY] - E[X]E[Y] = \frac{8}{3} - \frac{2}{3} \cdot 4 = 0$$

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y} = 0$$

Hence X and Y are uncorrelated.

Problem 3. A book has n typos. Two proofreaders, Sarah and Hannah, independently read the book. Sarah catches each typo with probability p_1 and misses it with probability $1 - p_1$, independently, and likewise for Hannah, who has probabilities p_2 of catching and $1 - p_2$ of missing each typo. Let X_1 be the number of typos caught by Sarah, X_2 be the number caught by Hannah, and X be the number caught by at least one of the two proofreaders.

- (a) Find the distribution of X .
- (b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$

(a) Define the following events:

S_i = typo i is found by Sarah

H_i = typo i is found by Hannah

$S_i \cup H_i$ = typo i is found by at least one of the proof readers

$$p = P(S_i \cup H_i) = P(S_i) + P(H_i) - P(S_i \cap H_i) = p_1 + p_2 - p_1 p_2$$

X = number of typos that is found by at least one of the proof readers $X \sim \text{Bin}(n, p)$

$$P(X = k) = \binom{n}{k} (p_1 + p_2 - p_1 p_2)^k (1 - (p_1 + p_2 - p_1 p_2))^{n-k}$$

(b) This can be solved using other method. I am writing the shortest proof.

$X_1 \sim \text{Bin}(n, p)$, hence, $X_1 = I_1 + I_2 + \dots + I_n$ where each $I_i \sim \text{Bern}(p)$ and I_i are independent.

$X_2 \sim \text{Bin}(n, p)$, hence, $X_2 = J_1 + J_2 + \dots + J_n$ where each $J_i \sim \text{Bern}(p)$ and J_i are independent.

X_1 and X_2 are also independent, hence,

$X_1 + X_2 = I_1 + I_2 + \dots + I_n + J_1 + J_2 + \dots + J_n$, meaning that $X_1 + X_2$ is sum of $2n$ independent Bernoulli trials where probability of success in each is p . Hence, by definition $X_1 + X_2 \sim \text{Bin}(2n, p)$.

Note that if $p_1 \neq p_2$, then this statement is not true, meaning even if X_1 and X_2 are independent Binomial, then we cannot claim $X_1 + X_2$ is also binomial.

$$P(X_1 = k | X_1 + X_2 = t) = \frac{P(X_1 = k, X_2 = t - k)}{P(X_1 + X_2 = t)} = \frac{P(X_1 = k)P(X_2 = t - k)}{P(X_1 + X_2 = t)} \quad (5)$$

$$= \frac{\binom{n}{k} p^k (1-p)^{n-k} \binom{n}{t-k} p^{t-k} (1-p)^{n-(t-k)}}{\binom{2n}{t} p^t (1-p)^{2n-t}} \quad (6)$$

$$= \frac{\binom{n}{k} \binom{n}{t-k}}{\binom{2n}{t}} \quad ; k = 0, 1, \dots, t \quad (7)$$

Problem 4. Let X be the number of Heads in 10 fair coin tosses.

- (a) Find the conditional PMF of X , given that the first two tosses both land Heads.
- (b) Find the conditional PMF of X , given that at least two tosses land Heads.

$$P(X = K | \text{the first two tosses land heads}) = \begin{cases} \frac{\binom{8}{k-2} (\frac{1}{2})^{10}}{(\frac{1}{2})^2} & k = 2, 3, \dots, 10 \\ 0 & k = 0, 1 \end{cases} \quad (8)$$

$$P(X = K | \text{the first two tosses land heads}) = \begin{cases} \binom{8}{k-2} (\frac{1}{2})^8 & k = 2, 3, \dots, 10 \\ 0 & k = 0, 1 \end{cases} \quad (9)$$

(b) E = at least two tosses land heads

$$P(X = K|E) = \begin{cases} \frac{\binom{10}{k}(\frac{1}{2})^{10}}{1 - ((\frac{1}{2})^{10} + \binom{10}{1}(\frac{1}{2})^{10})} & k = 2, 3, \dots, 10 \\ 0 & k = 0, 1 \end{cases} \quad (10)$$

$$P(X = K|E) = \begin{cases} \frac{\binom{10}{k}}{2^{10} - 11} & k = 2, 3, \dots, 10 \\ 0 & k = 0, 1 \end{cases} \quad (11)$$

Problem 5. Let X and Y be independent random variables following Poisson distributions with parameters λ_1 and λ_2 . Find the distribution of $Z = X + Y$.

Since $X \sim Pois(\lambda_1)$, for any i , we have $P(X = i) = e^{-\lambda_1} \frac{\lambda_1^i}{i!}$

Since $Y \sim Pois(\lambda_2)$, for any j , we have $P(Y = j) = e^{-\lambda_2} \frac{\lambda_2^j}{j!}$

$$P(X + Y = k) = \sum_{i=0}^k P(X = i, Y = k - i)$$

Since X and Y are independent, we can write $P(X = i, Y = k - i) = P(X = i)P(Y = k - i)$. Hence,

$$P(X + Y = k) = \sum_{i=0}^k P(X = i)P(Y = k - i)$$

$$P(X + Y = k) = e^{-\lambda_1 - \lambda_2} \sum_{i=0}^k \frac{\lambda_1^i}{i!} \frac{\lambda_2^{k-i}}{(k-i)!} = e^{-\lambda_1 - \lambda_2} \sum_{i=0}^k \frac{\lambda_1^i}{i!} \frac{\lambda_2^{k-i}}{(k-i)!} \frac{k!}{k!} = \frac{1}{k!} e^{-\lambda_1 - \lambda_2} \sum_{i=0}^k \frac{k!}{i!(k-i)!} \lambda_1^i \lambda_2^{k-i}$$

We know $\binom{k}{i} = \frac{k!}{i!(k-i)!}$ and by binomial theorem $\sum_{i=0}^k \binom{k}{i} \lambda_1^i \lambda_2^{k-i} = (\lambda_1 + \lambda_2)^k$, hence,

$$P(X + Y = k) = \frac{(\lambda_1 + \lambda_2)^k}{k!} e^{-(\lambda_1 + \lambda_2)}$$

Hence, $X + Y \sim Pois(\lambda_1 + \lambda_2)$

Problem 6. Each of $n \geq 2$ people puts his or her name on a slip of paper (no two have the same name). The slips of paper are shuffled in a hat, and then each person draws one

(uniformly at random at each stage, without replacement). Find the mean and standard deviation of the number of people who draw their own names.

X = number of people who draw their own hat

$X = I_1 + I_2 + \cdots + I_n$ where

$$I_i = \begin{cases} 1 & \text{if a person } i \text{ draws their own hat} \\ 0 & \text{otherwise} \end{cases}$$

$$P(I_i = 1) = \frac{1}{n}.$$

$$E[X] = \sum_{i=1}^n E[I_i] = n \cdot \frac{1}{n} = 1$$

$$\begin{aligned} \text{Var}(X) &= \text{Cov}(X, X) = \text{Cov}\left(\sum_{i=1}^n I_i, \sum_{j=1}^n I_j\right) = \sum_{j=1}^n \text{Var}(I_i) + \sum_{i \neq j} \text{Cov}(I_i, I_j) \\ &= n \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right) + (n^2 - n)(E[I_1 I_2] - E[I_1]E[I_2]) \end{aligned}$$

$$I_1 I_2 = \begin{cases} 1 & \text{if } I_1 = 1 \text{ and } I_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$P(I_1 = 1, I_2 = 1) = \frac{1}{n} \cdot \frac{1}{n-1} = \frac{1}{n(n-1)}$$

$$\begin{aligned} \text{Var}(X) &= 1 - \frac{1}{n} + (n^2 - n) \left(\frac{1}{n(n-1)} - \frac{1}{n^2} \right) \\ &= 1 - \frac{1}{n} + 1 - \frac{n-1}{n} = 1 \end{aligned}$$